

Article

Exploring behavior patterns in human and machine interactions

Ping Zhao^{a,b,1}, Yunjie Wei^{a,c,1,*}, Shouyang Wang^{a,c,1}^a Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing 100190, China^b School of Mathematical Sciences, University of Chinese Academy of Sciences, Beijing 100190, China^c Center for Forecasting Science, Chinese Academy of Sciences, Beijing 100190, China

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ABSTRACT

A framework based on response-dependence theory is proposed to analyze human and machine interaction patterns. The interaction is an information processing and exchanging system with agents responding to signals from the decision perspective on multiple channels. Agents' pattern differences may emerge via a multiple-input multiple-output model on the sequential machine data. The connectivity with different decision-making factors embodies the interactive narrative features. The magnitude of the frequency response is compared in a parallel analysis. The interaction relationship will emerge in frequency response comparison. In the empirical study, we compare AlphaGo and human professional players' behaviors in the game of Go. We find a similar connectivity structure in vertical analysis and coincidences with the interaction relationship in the game. However, the response magnitude interval differs between AlphaGo and human professional players in several stimulus-response pairs. Integrated with static comparison, we find AlphaGo is more sensitive to long-term payoff changes than human professional players. The framework and empirical studies indicate that theory and techniques from cross-disciplinary can provide a perceptive and objective explanation of behavior patterns in human and machine interaction. The proposed framework might benefit scientists in research on AI ethics and machine behavior interpretation with big data techniques.

1. Introduction

With the rapid development of the techniques of AI, intelligent machines have played a vital role in various fields of human activities. To recognize human behavior patterns is one of the scientists' focuses, for example, facial expressions in understanding human emotions [1,2], conversation [3], body language [4], or the relationship between behaviors and neuro-scientific measurements such as electroencephalogram [5]. Scientists have greatly improved AI performance in human-machine interaction (HMI) and noted some problems. For example, human expressions might not reflect the real emotion of the subject [1]. The neuro measurement can indicate human behaviors [6] but often needs to be analyzed with elaborately designed events and might not easily be explained. Furthermore, the neuro approach can hardly apply to AIs. Scientists focus more on results provided by AI in understanding humans without considering the activities of AI itself. Many existing researches concentrate on external activities or indirect reflection of human activities. And we still need a perceivable and interpretable approach to connect the inner drive to human behaviors. For social computing purposes [7], we need a finer characterization of HMI. Besides, scientists expect the studies on HMI will help the cooperation between machines

and humans in more areas [8]. Therefore, we need to explore and extend machine behavior studies to inner activity analysis for both humans and AI and to improve the AI techniques with caution so that we try to minimize the negative impact on humans as much as possible.

Our studies are motivated by the recent discussion about the imitation game [9] from a philosophical angle [10]. Philosophers provide interesting insights from different angles about the imitation game. The main question is whether we need to test machine behavior to infer that the computer's behavior is appropriately human-like compared to the "average" human interrogator in a canonical statement. If it is yes, how do we achieve it? In other words, how can we make intelligent machines perceive the difference in behavior patterns from other agents in interactions?

Testing machine behavior is necessary. Machine behavior analysis helps scientists better understand the behavior patterns in HMI. In the canonical understanding of the Turing test, 'average' human interrogators score on the agents. Turing assumes the interrogator is 'highly competent' but 'forbidden any detailed knowledge of the machine's architectures. Though the "average" interrogator can provide intuitive information about interactions, the judgment might be limited to some predefined areas and short of accuracy. Not to mention the bias of the

* Corresponding author.

E-mail address: weiyunjie@amss.ac.cn (Y. Wei).¹ These authors contributed equally to this work.

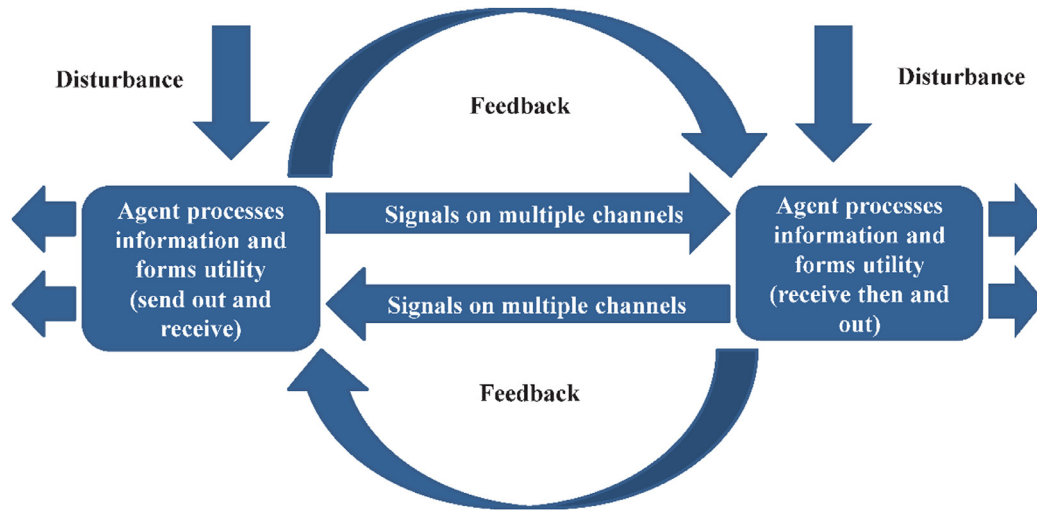


Fig. 1. Information exchange in interaction.

interrogator is always an unsettled dispute. Since we expect machines to play a better role in humanized interaction scenes like health care, communications, and education, we need to ensure that they are competent in the areas of human concern. Then an analysis of the algorithms and data of the AI may help fill the gaps in understanding both human and machine behaviors. In essential, to improve machine performance in HMI, we need to integrate the advantages of both humans and machines. Conventional approaches like expression or neuro measurements work on human behavior but might be inapplicable to machine behavior analysis. Recently, scientists have proposed a comprehensive framework for machine behavior research [11,12], which helps light the direction of human and machine interaction research at a high level. Scientists in machine behavior studies try to integrate algorithms and data analysis with cross-disciplinary knowledge and techniques to interpret machine behavior. In particular, analyzing data-bearing decision meaning can provide a perceptive and objective explanation of behavior patterns.

We suggest considering HMI as a complex system [13] with agents that exchange various types of information regarding their concerns, as Fig. 1 displays. The interaction comprises information reception (or signal input), a utility-generating process, and information dispatch (or signal output). The utility-generating process relates to how agents shape their utility with the information. We can separate it from the rest of the system. For behavior pattern discernment, analyzing the input and output signals might suffice for the response-dependence theory examination proposed for imitation games [9,10,14,15]. Scientists have applied the response-dependence theory in neuropsychology to identify specific cognitive functions by testing agents' responses to controlled stimuli [16,17]. Moreover, the study found that consumer behaviors toward input information like advertisements are also a response-dependent analysis [18].

We propose a framework based on the response-dependence theory and the imitation game, incorporating cross-disciplinary theories and techniques to conduct behavior pattern discernment. First, the principle is to allow behavior patterns to emerge from the comparison of machine-machine interactions, human-machine interactions, and human-human interactions by applying data-driven approaches to AI records of interactions. The aim is to capture sufficient information for comparing agents' behavior patterns under an objective and comparative background considering agents' flexible and adaptive responses. Meanwhile, to ensure the comparability of data across multiple factors, it is necessary to consider data integration and separation in building response systems based on the experiments. For example, in this paper, we differentiate between a 6-variable and a 4-variable system, as two variables in the 6-variable system can integrate into one variable in the 4-variable system. A sep-

arate analysis is necessary since both systems can be explained from the behavioral perspective. Then, the pillar of the framework is the comprehensive synthesis of the leading knowledge from decision science, information science, and physical implications, creating a solid ground for behavior interpretations. Next, we embody the framework with three research questions discussing behavior pattern discernments. The three questions can be easily understood following the logic of mechanical analysis, an area studying physical interactions by systematically discussing forces. The three distinguishing features in studying forces are the point of application, magnitude, and direction. In the framework, RQ1 is about identifying the "point of application" in interactions. We suggest identifying the primary connections between multiple factors based on the understanding of the narrative nature of sequential decision-making. RQ2 is about the examination of magnitude. We propose to conduct a paired T-test of both players' response magnitudes on a unified frequency band. RQ3 is about direction identification. We suggest using physical interpretations on the interactive relationships of outcome change sequences. In summary, the empirical analysis based on the three RQs can be conducted through a multiple-input and multiple-output (MIMO) approach to model the parts of information receiving and dispatching in agent interactions.

We suggest examining the agent behaviors for some specific lengths instead of at some point. The reason is that when we relax the observation to some specific actions or periods, we consider the context. Under a similar situation, agent behaviors might have been led by the nature or habits of the decision-maker or shaped by the social norms in interactions. The agent behavior is not necessarily the same in the context, but the behavior is similar within some ranges. The interrogator might be unable to tell the difference when the behavior is within the range but can distinguish when it is across the boundary. We suggest the HMI study can focus on the change of the factors bearing decisive meanings and reflecting the impacts of the inner activities of agents, i.e., the gain and loss sequences of concerned factors.

In our empirical studies based on the game of Go, we study the decision payoff sequences between AlphaGo and human professional players with multivariate techniques in two different decision modes. First, we collect data from AI bots regarding players' behaviors. Furthermore, we preprocess the data to reflect the inner activities of agents in decision-making. Then we conduct a static, dynamic comparison of players' behavior patterns, followed by an analysis of the interaction relationships. Besides conventional descriptive statistics analysis, we suggest including spectral analysis to compare the cyclical features of players' intertemporal decisions. In dynamic comparisons, we employ spectral pairwise-conditional granger causality and frequency response to calibrate the

stimulus-response pairs within the frequency domain. The frequency response approach also represents the players' interaction relationship.

The primary contributions are: First, we propose an operational framework for behavior pattern identification in HMI. There are few studies discussing relevant theory and methodology grounds systematically and comprehensively. The framework in this paper synthesizes knowledge from decision and information areas and provides a general background of HMI analysis regarding agents' inner activities. Second, we introduce signal analysis methodologies for HMI empirical analysis to meet the needs of the interaction process analysis. The approaches in this paper help identify the difference in agent behavior patterns and reflect agent relationships during interactions. Moreover, it may further help scientists understand and control the impacts of AI on humans. Finally, we extend the new directions of machine behavior in HMI studies, which may help promote studies in the boundary of rationality, mechanism of emotions, and quantitative analysis of AI ethics in the future.

The rest of the paper is arranged as follows. Section II raises research questions from different perspectives and briefly introduces methods and data. In section III, we illustrate the empirical results from both static and dynamic perspectives. Then, we summarize our studies, discuss limits and future directions, and finally conclude.

2. Problems and methods

2.1. Research questions in HMI

Considering HMI as a complex system, we formulate three research questions to study HMI from the logic of mechanical analysis, which also applies to machine-machine and human-human interactions. The research questions include the individual patterns from a canonical research point of view and interactive patterns from a comprehensive view, incorporating the stimuli-response structure, reaction response magnitude, and interactive relationships or directions. The questions are created for a systemic and quantitative study on interactions, which might further contribute to AI ethics and crossing areas.

2.1.1. The narrative nature of sequential decision-making

Studying the narrative nature of sequential decision-making may shed light on the statistical learning mechanism from the interaction background. Sequential decision-making or intertemporal decision-making refers to the sequential decisions of an agent over some periods. In canonical decision theories, the utility results derive from a decision governed by a discount factor in much the same way that financial markets evaluate losses and gains [19]. The narratives of intertemporal decisions are framed from a time perspective, and the outcome at each sampling time point internally connects with each other. Agreements have arrived that utility changes can be evaluated, predicted, or even reflected in humans' short-term emotional change [20]. Furthermore, scientists have worked on human sentiment patterns based on big data [21] or even conducted AI emotion computation grounded in decisions [22].

In the physical interpretation, sequential decision-making can be understood as a motion process, and the gain & loss, or the change of utility sequences related to each decision, is the speed of sequential outcomes or values [23]. Physical approaches might simulate the periodic patterns of utility oscillation. In particular, for strategic decision-makers, changes are in plans, and a certain degree of gain & loss is acceptable and does not affect their behavior patterns. However, the narrative feature of the physical explanation might be more indirect. For example, changes might be observed when the outcome is apparent instead of at each step. It needs some physical methods to reveal the connections in sequential decisions.

In the interaction context, the agent's decisions might depend not only on the agent's internal utility-generating process but also on the behavior of others. We consider that the connection between different agents' behavior outcomes reflects the interactive narrative features.

Suppose the results of the subsequent behaviors of an agent are predictable by other agents. In that case, we can assume predictability as a "causality" measure of the narrative features in the interactions. For instance, a player must consider the opponent's decisions at every move in board games. In social interactions, the narrative patterns between agents may reflect the spirits behind social norms or codes of conduct for different cultures.

The narrative nature of interactive behaviors helps to identify the "point of application," i.e., the primary connections in interactions involving multiple factors. Therefore, the first research question is as follows.

RQ1. What are the individual narrative patterns of AI and humans in intertemporal decisions? Furthermore, what are the "causal" patterns between different signals for AI and humans?

2.1.2. Response magnitude in interactions

The comparison of response magnitude between AI and humans helps understand AI alignment, which is essential in building an appropriate AI ethics framework. During interactions with humans, besides continuously improving performance, AI is expected to be able to learn and flexibly comply with social values, cultures, and codes. In other words, AI should behave in an agreed results that does not conflict with human moral beliefs or values [24]. Therefore, AI must gradually become aware of how to respond to the counterparty, especially the boundary of responses. However, the response guidelines and corresponding boundary lines might vary on the various interactive contexts and counterparts. Under a dynamic social background, we suggest that the response guidelines and the boundary lines can be mapped to the agent's perception of the interaction. As Adam Smith states in 'The Theory of Moral Sentiments',

"Every faculty in one man is the measure by which he judges of the like faculty in another. I judge of your sight by my sight, of your ear by my ear, of your reason by my reason, of your resentment by my resentment, of your love by my love. I neither have, nor can have, any other way of judging about them."

We assume agents are embedded with similar behavior patterns which arise from intuitions, knowledge, or experiences in the interactions and learning process. Agents' responses may be in various forms, but usually, the magnitude of response is likely to be drifted in some tacit ranges for specific contexts. Statistical approaches like paired tests can detect over-reaction and under-reaction over the response magnitude boundaries.

RQ2. How do AI and humans respond to different signals?

2.1.3. Social dilemma analysis in HMI

Understanding social dilemmas is essential to avoid potential harnesses caused by AI and critical to direct the HMI to be safe and beneficial to humans in real life. A social dilemma is broadly defined as a situation in which short-term self-interest is at odds with longer-term collective interests [25,26]. Psychologists summarized social dilemmas to theories such as reward and punishment, resource and benefits asymmetries, and perception of uncertainty [26]. Research about social dilemmas often connects to environmental subjects [27,28]. Studies on the social dilemma with collective narratives in group behavior also attract attention [29]. Recently, scientists have examined the impact of information and the role difference in agents' behaviors in sequential social dilemma games [30,31]. The studies on social dilemmas may shed light on the research on AI ethics. To promote the application of intelligent machines in society, scientists aim to develop techniques that ensure AI could cooperate with humans instead of harming them. Hence it is essential to quantify the interactive relationship and set the measurement as an important factor in decision-making. The studies on social dilemmas are usually based on game analysis and evolutionary techniques. Extending the theories and methods of social dilemma to the algorithm with big data is vital in the future. By considering sequential decisions

as a motion process, it is easy to examine whether the outcomes of decisions between agents are in the same direction. When the physical measure of paired outcome changes aligns with the range of co-movement direction, the interests of agents are non-conflict or even in mutual benefits. Otherwise, the agents are generally conflicted, which might lead to a social dilemma.

RQ3. What are the patterns of social dilemmas in interactions?

2.2. Methods

For the static narrative patterns of individuals in RQ1, i.e., the potential cyclical patterns of payoff changes, we follow the existing literature [23] and use spectral analysis, a canonical methodology in signals analysis in the frequency domain. In particular, multitaper spectral analysis is suitable for finite sequences in extracting the significant periods in oscillated signals with the F-ratio test. Sequences for spectral analysis need to be stable, and we conduct ADF tests to ensure the prerequisite is fulfilled. The static comparison of the periods' distribution density for different agents which we discuss for the RQ1 is conducted with the multitaper package in R² [32].

We define interactive narrative patterns in RQ1 as the predictability of an agent's behavior toward others. Granger causality is a canonical approach in testing the predictability of a variable at some significance levels. Hence it can depict the vertical "causality" and "connectivity" in a system comprised of sequential decision signals and uncover the main interactive structure in complex systems [33,34], which is like the "point of application" in studying forces. AI techniques naturally embed the capacity of predictability. They have been widely used and recognized in various areas of forecasting, like tourism and financial instruments, which are usually analyzed in an integrated way [35,36]. This paper applies multivariable techniques, i.e., the MVGC toolbox [37], for static and dynamic interactive narrative pattern identifications. Furthermore, we conduct frequency responses and make a Bode plot with the System Identification Toolbox in Matlab.

Regarding RQ2 of response magnitude comparison, we apply the frequency responses analysis based on the MIMO approach [38], which transfers the agent response to an identical frequency band (0–0.5 Hz in our study) for easy comparison. After that, a paired T-test is conducted on the lower and upper boundary within the 95% confidential interval of the response magnitudes between the two players in each type of game.

Since the interactive relationship cannot be observed directly, RQ3 of social dilemma analysis is conducted via a data-driven approach by summarizing the phase coherence within each type of games [23], in which statistical patterns emerge directly. Moreover, phase coherence is one of the direct results of the Bode plot.

We generally apply multivariate approaches to examine the three research questions and reveal comprehensive behavioral patterns. And we focus on results in the unified frequency bands. Please refer to the supplementary for more detail information about the methods.

2.3. Data and measure

In this paper, we take Go as an example [39], as the game is laboratory-like, and relevant data is easy to access. In the game of Go, both players compete in the same condition without any disturbances. Since the contexts and outcomes of each decision depend on the previous situation on the board under a clear valuation rule, if the AI bot is well trained and has a persuasive history in the games with top human players, the bot might reflect quite close results as when the game was played.

² Please refer to K. J. Rahim's Doctoral dissertation for more details about the R package. Applications of Multitaper Spectral Analysis to Nonstationary Data. Queen's University (2014).

We follow the previous study [23] and examine three different types of games: Type A - AlphaGo Master (AGM) vs. AlphaGo Zero 40 blocks of the final version (AGZ), interactions between different versions of AI; Type B - AlphaGo Master vs. human professional players (Pros), interactions between AI and humans; and Type C - rivals vs. 7 human professional players (7 pros), interactions between human professional players. The three types of games represent interactions between different agents, constructing a coherent and comparable experiment in deeply understanding HMI [12]. We also construct a sub-sample of Type B - AlphaGo Master vs. 7 pros for a better comparison. The 7 pros play with AlphaGo Master online or formally more than once. Considering pros might exhibit different patterns when playing with AlphaGo Master compared to game against other pros, we include 84 games where the 7 pros played with other pros just before they played online games with AlphaGo Master within six months. Moreover, there are Go games between different versions of AlphaGo and pros, for example, AlphaGo Fan vs. Fan Hui, AlphaGo Lee vs. Lee Sedol, and games between different versions of AlphaGo like AlphaGo Master vs. AlphaGo Lee [23]. However, since the games are insufficient for statistical analysis, we focus on the games of AlphaGo Master in the study.

We extract relevant data from the review of Jueyi or FineArt, an AI bot deployed on the Tencent game platform, which has been widely seen as the 2nd most vital AI after AlphaGo. Though we are unaware of the detailed architecture of FineArt, the output of FineArt includes both short-term and long-term payoff of winning rate for player's every decision, similar to AlphaGo. Considering that most AI conducts massive calculations on the Monte Carlo tree search, the assessment of each move might be close between FineArt and AlphaGo. In addition, AlphaGo Master and AlphaGo Zero have the same neural network structure but different training approaches. AlphaGo Master is trained with existing games between professional players however, AlphaGo Zero acquires skills in Go purely from self-learning and without human knowledge. The output of AlphaGo Master and AlphaGo Zero includes a move probability representing the probability of selecting each move, which can be a proxy reflecting the complexity of the decision. At the same time, the moving probability is a subjective variable that might vary between AI and pros as it is influenced by individual experience and the AI's programming. Even for pros, their judgments might be different. In this paper, move probability-relevant data is absent. Another output is the estimation of the probability of the current player winning from the position, also known as the winning rate, a proxy reflecting the decision's long-term consequence or temporal construal. Though the perception of the winning rate might differ for players in the same situation, the Monte Carlo Tree search still provides a relevant objective measurement. Finally, we take the payoff assessment by each move as a proxy of the short-term consequence of the decision. By comparing to the existing study of intertemporal decisions from the game of Go [23], we extend the comparison to only the short-term payoff sequences and to a more comprehensive data structure, including different types of sequences more closely related to real life.

The multiple payoff sequences direct our analysis to two different types of multivariable systems: the 6-variable MIMO and the 4-variable MIMO, which the former system can be treated as an extension of some variable in the latter one. MIMO is usually applied in communicational frameworks and is noble in analyzing human and computer behaviors. Both types contain a winning rate, a proxy from construal-level theory about temporal distance changes that reflect people's responses to future events by changing how people mentally represent those events [40]. Moreover, the 6-variable MIMO separates the payoff of the agent herself from the payoff to the opponent. In contrast, the 4-variable MIMO integrates the payoff by considering the net payoff of the agent in considering possible mental accounting effects, which often refers to how people mentally sort money into different accounts in consumptions [41]. Besides, since the game of Go is a typical sequential game with two players placing their stones in turns, there are two sub-systems where the player and the opponent act as responders, respectively. Please refer to part C

Table 1
Game type, implication, variables, and the indications.^a

Game type	Implication of the game type	Proxy (in all types of games)	Variables - 1st order difference of the proxy (abbreviation)
Type A^b Games between AI of different versions (20 games)	To compare decision patterns of AI with different learning approaches.	Agent payoff	Short-term proceeds (STPs)
Type B Games between AI and humans (63 games)	To compare decision patterns from AI and humans.	Opponent payoff impacted by agent decision	STPs-oppo
Type B sub-sample^c Games between AI and specific humans (26 games)	To construct a sub-sample for specific human players for the sake of coherent comparison to human-human games.	Net payoff (agent payoff-opponent payoff)	Net proceeds (NPs)
Type C^d Games between humans (84 games)	To compare the decision patterns of human professional players.	Winning rate	Long-term proceeds (LTPs)

^a Stimulus is always the input signal, and the responder always refers to the output signal. STPs, STPs-oppo only appear in 6-variable MIMO. NPs only appear in 4-variable MIMO. LTPs appear in both 6-variable and 4-variable MIMO.

^b AlphaGo Master (AGM) is trained based on existing human game records. AlphaGo Zero 40 blocks final version (AGZ) self-learns the game of Go without human knowledge.

^c The sub-sample of Type B consists of the 7 pros who play with AlphaGo Master online or formally more than once.

^d Type C consists of games of the 7 pros with other human professional players (rivals).

in the supplementary, which illustrates the data structure of different sub-systems.

Our analysis focuses on the first order of difference for all proxies to avoid the impacts of different rules. The game types, proxy, and the indication of its first-order differences (the gain and loss sequences) are elaborated in Table 1. We conduct ADF tests on all gain and loss sequences for further spectral analysis. Moreover, all is stationary on a 5% significance level (see supplementary).

3. Results

3.1. Descriptive statistics

In general, Fig. 2 shows different patterns between short-term and long-term result changes, particularly in the indices of mean and skewness. The reason might relate to the competitive nature of the long-term result, i.e., winning rate, for which the two players are in a consistent competitive connection, and one player’s change is opposite to the opponent’s value. There is no noticeable difference in the kurtosis index for all agents (panel f).

Regarding different types of games, there are some differences in the indices of Min., Max, and Standard deviation (panel a, b, and e) between humans and AlphaGo. Additionally, AGZ is better than AGM in long-term proceeds, particularly in indices of mean and skewness in Type A games (panel c and d). For Type B games, AGM is better than pros and the 7 pros in long-term proceeds in indices of mean and skewness (panels c and d) and net proceeds in the index of the mean (panel c). The computing capacity might lead to AIs’ advantage on the competitive indices. While pros’ mistakes might also result in AI leading the game, it can imply humans might be less sensitive in some measures than AGM, mainly the long-term proceeds. There is no apparent difference for players in Type C games.

3.2. RQ1 individual static narrative patterns

In canonical analysis, individual decision patterns are reflected by a utility function with a discounting factor. Instead, we suggest making the individual patterns emerge by data mining approach on the 1st order of proxies. Specifically, we examine the significant periods with the same approaches and parameters as previous studies [23]. Players will consider a specific length when making strategic decisions in a limited resource competition game. Hence the gain and loss sequences may have some cyclical patterns. We take the sampling frequency as 1 Hz for each move of the focal player. We capture the significant periods on all proxies (see Fig 3) based on the F-ratio test in multitaper spectral analysis [32] for a static comparison. The most obvious difference

is the periods on LTPs. AGM and pros display a flat pattern on long-term proceeds in Type B games. However, in Type C games, pros and rivals still exhibit a vertical pattern in LTPs. By comparing a limited sample of Type A games, we note that the two AIs show different patterns on proxies of long-term (LTPs) and net proceeds (NPs). The difference in LTPs pattern between AI and pros implies that humans might be insensitive to long-term change compared to AI.

To summarize, we note significant differences in the narrative patterns between AI and human professional players in the individual static narrative pattern comparison of RQ1, especially in the long-term result changes. Human players might be less sensitive to the winning rate changes than AlphaGo.

3.3. RQ1 interactive narrative patterns – the vertical comparison

People usually have to consider multiple variables when making decisions. Therefore, it is necessary to investigate the general connectivity among different variables of concern in the multivariable decision system in the interactive background. The predictability of the responses from pros and machines, based on various types of stimuli, can effectively reveal the underlying connections through vertical analysis using data mining techniques.

We calculate pairwise-conditional Granger causality for different types of systems and summarize the proportion of significant connectivity passing the permutation test for each pair. The proportion implies the directional predictability in the stimulus-response relationship for input and output signals. From Table 2 and Table 3, we note similar patterns in general connectivity for all players, i.e., the bi-directions between STPs and STPs-oppo and LTPs-stimulus to LTPs-responder of Table 2 for the 6-variable MIMO, and NPs-stimulus to NPs-responder and LTPs-stimulus to LTPs- responder of Table 3 for the 4-variable MIMO.

We recognize some differences for players in different kinds of games as well. In Table 2 for the 6-variable MIMO, the forecasted patterns of the three main stimulus-responses in bold of AIs are similar in Type A games (35.00% vs. 40.00%, 50.00% vs. 55.00%, 60.00% vs. 60.00%). The predictability difference between the two short-term pairs of AGM is not apparent when comparing pros (25.81% vs. 29.03%, 22.58% vs. 19.35%) or the 7 pros (40.00% vs. 36.00%, 28.00% vs. 32.00%) in Type B games. However, pros (7 pros) display higher predictability in LTPs than AGM (91.94% vs. 22.58% for pros vs. AGM, 92.00% vs. 48.00% for 7 pros vs. AGM). In Type C games, the forecasted patterns of LTPs response are higher than the ones of short-term outcomes response but similar between human players (75.00% vs. 67.86% for LTPs, 19.05% vs. 22.62% and 28.57% vs. 23.81% between STPs and STPs-oppo).

In Table 3 for the 4-variable MIMO, the response in bold of AGM is more predictable than the AGZ (65.00% vs. 45.00% for NPs, 80.00%

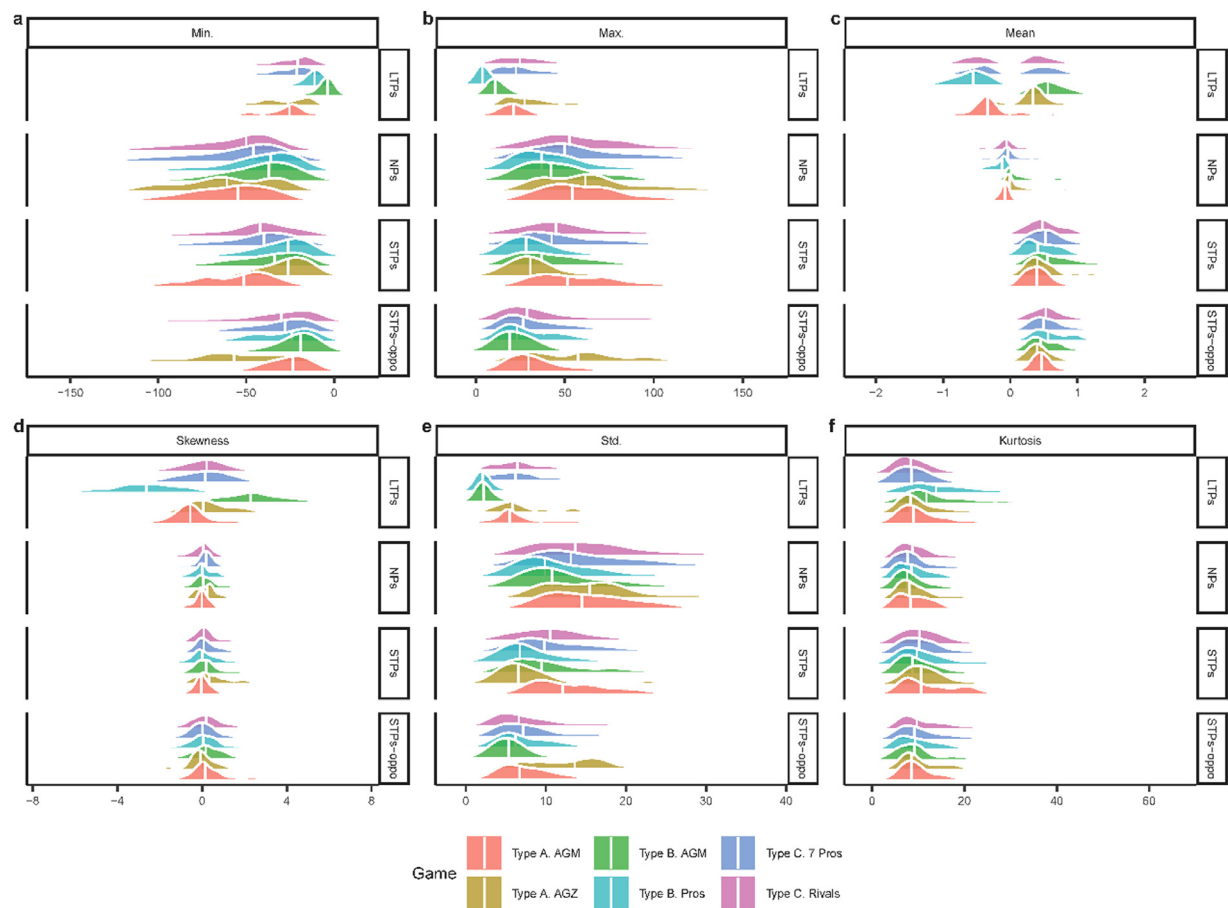


Fig. 2. Descriptive statistics.
Note: Game refers to game type and the responder; for example, Type A. AGM indicates AGM as the responder in Type A. games. The white vertical line in the ridge is the median of the corresponding group.

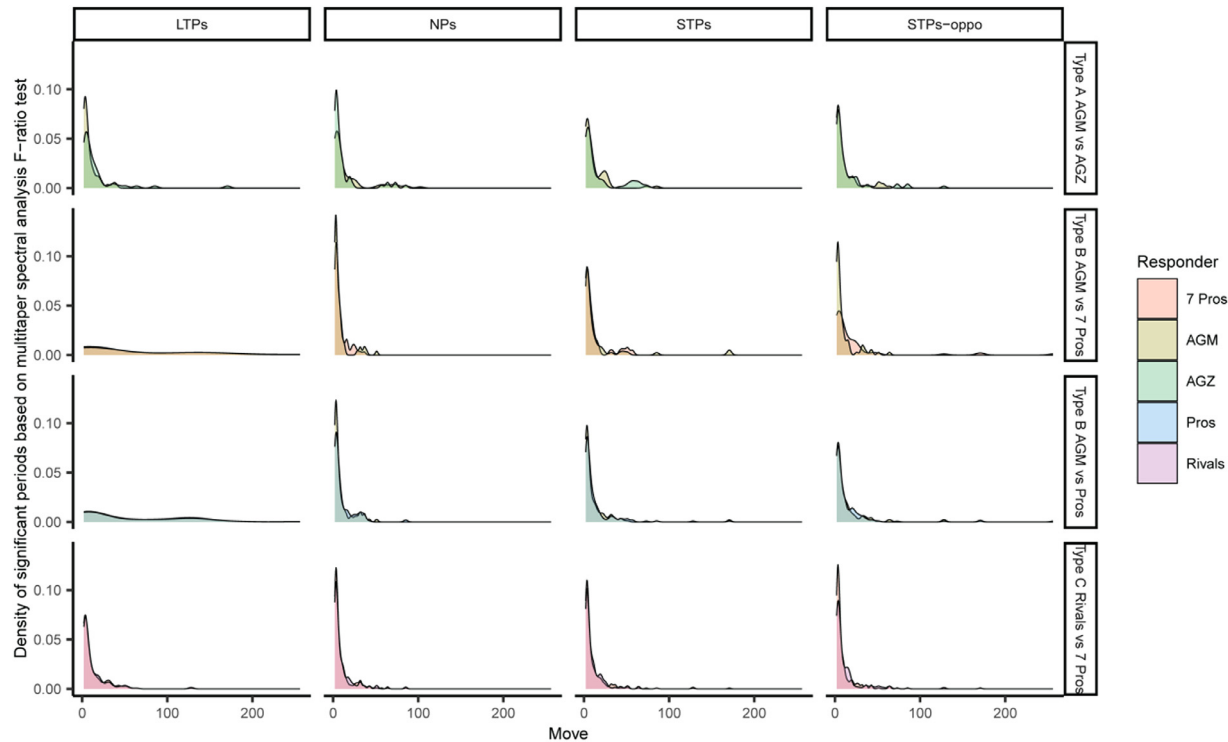


Fig. 3. The distribution density of significant periods. Parameters in multitaper spectral analysis: $NW=4$, $K = 2NW-1 = 7$. The significance level of the F-stats test: 5%.

Table 2
Pairwise-conditional Granger causality proportion comparison for 6-variable MIMO.

Pairs/Game & responder		Type A		Type B		Type B-sub sample		Type C	
Stimulus	Response	AGM	AGZ	AGM	Pros	AGM	7 Pros	Rivals	7 Pros
STPs	STPs	10.00%	0.00%	3.23%	1.61%	0.00%	0.00%	1.19%	1.19%
STPs	STPs-oppo	35.00%	40.00%	25.81%	29.03%	40.00%	36.00%	19.05%	22.62%
STPs	LTPs	0.00%	0.00%	3.23%	0.00%	4.00%	0.00%	0.00%	2.38%
STPs-oppo	STPs	50.00%	55.00%	22.58%	19.35%	28.00%	32.00%	28.57%	23.81%
STPs-oppo	STPs-oppo	0.00%	0.00%	1.61%	3.23%	0.00%	4.00%	1.19%	0.00%
STPs-oppo	LTPs	5.00%	5.00%	0.00%	0.00%	0.00%	0.00%	0.00%	2.38%
LTPs	STPs	0.00%	0.00%	0.00%	6.45%	0.00%	4.00%	0.00%	1.19%
LTPs	STPs-oppo	0.00%	5.00%	0.00%	0.00%	0.00%	0.00%	1.19%	1.19%
LTPs	LTPs	60.00%	60.00%	22.58%	91.94%	48.00%	92.00%	75.00%	67.86%

Note: Significance level = 5%, number of permutations =1000.

Table 3
Pairwise-conditional Granger causality proportion comparison for 4-variable MIMO.

Pairs/Game & responder		Type A		Type B		Type B-sub sample		Type C	
Stimulus	Response	AGM	AGZ	AGM	Pros	AGM	7 Pros	Rivals	7 Pros
NPs	NPs	65.00%	45.00%	27.42%	29.03%	44.00%	36.00%	32.14%	30.95%
NPs	LTPs	5.00%	0.00%	0.00%	3.23%	0.00%	4.00%	1.19%	2.38%
LTPs	NPs	5.00%	5.00%	0.00%	3.23%	0.00%	0.00%	3.57%	2.38%
LTPs	LTPs	80.00%	60.00%	29.03%	93.55%	40.00%	92.00%	80.95%	80.95%

Note: Significance level = 5%, number of permutations =1000.

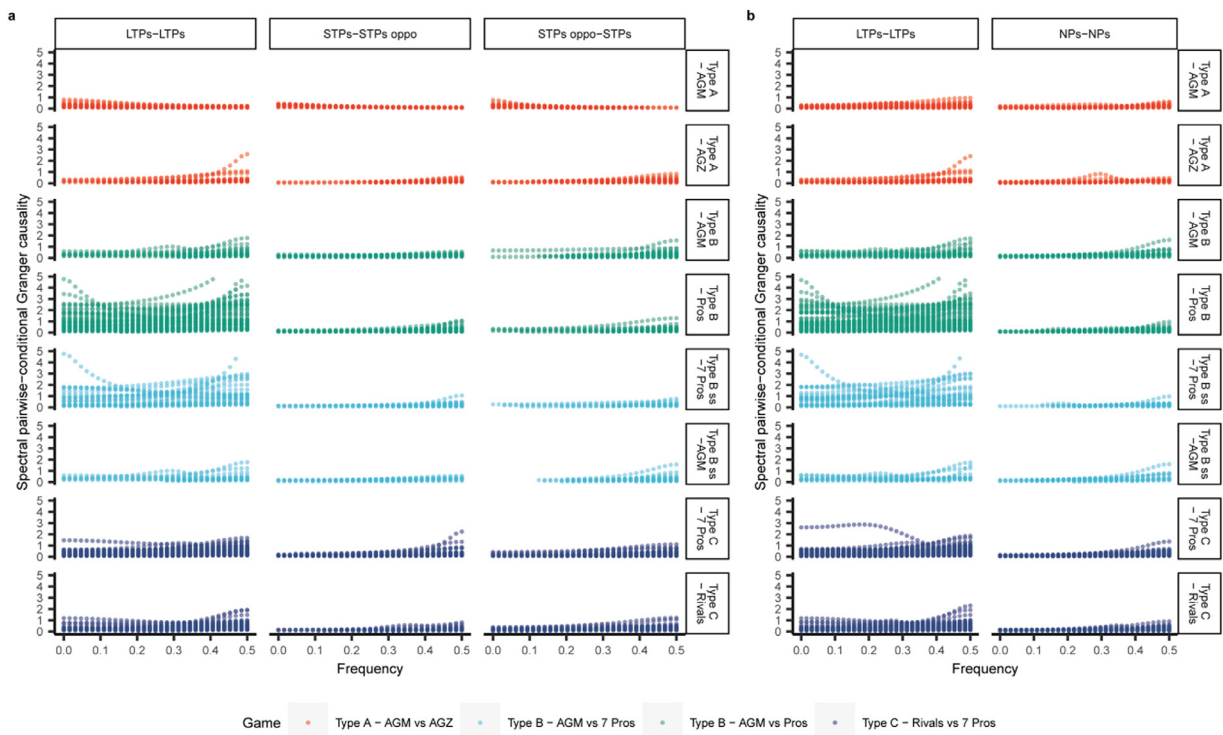


Fig. 4. Spectral pairwise-conditional Granger causality (spwcgc) proportion comparison for the main stimulus-response pairs of 6-variable MIMO (left) and 4-variable MIMO (right). The tag on the right side refers to the game-responder. Sample rate = 1 Hz, frequency resolution = 32, significance level = 5%, number of permutation =1000. We remove one game from the spwcgc consolidation (game code pmt25, please refer to the supplementary) as the VAR during estimation is unstable for further processing.

vs. 60.00% for LTPs). In Type B games, the difference in forecast patterns of NPs pair is less noticeable between AGM and pros, whereas the predictability of LTPs exhibits more variation. (29.03% vs. 93.55% for AGM vs. pros; 40.00% vs. 92.00% for AGM vs. 7 pros). Similar to the 6-variable MIMO, the forecasted patterns of LTPs response are higher than short-term outcomes but similar between human players in Type C games (32.14% vs. 30.95% for NPs, 80.95% vs. 80.95% for LTPs).

In Fig. 4, we compare the significant spectral pairwise-conditional Granger causality (spwcgc) values for the primary stimulus-response relationships. In the 6-variable MIMO, we observe the magnitude of the machine game is more even than that of humans. The spectral pairwise-conditional Granger causality magnitude on LTPs of human players looks greater than those on STPs and STPs-oppo for games between machine and pros (Type B games) and games between human players (Type C games). Meanwhile, the spectral pairwise-conditional Granger causal-

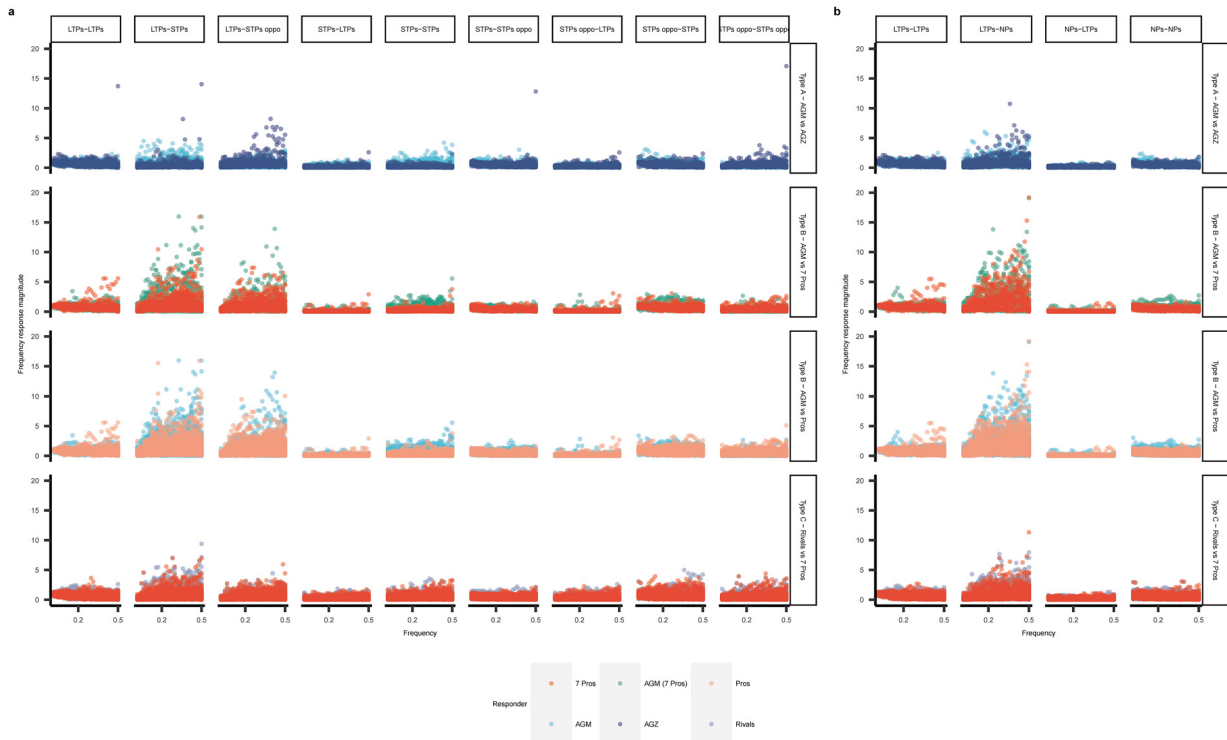


Fig. 5. Frequency response comparison for the 6-variable MIMO (left) and the 4-variable MIMO (right). The tag on the top is the stimulus-response pair. We use the default parameters of Syntax Spa in Matlab.

ity discovers that the magnitude of LTPs-LTPs of AGM response is less than the pros. In general, the LTPs of the AGM decision can predict the corresponding pros' response to some extent, while the LTPs of the pros can barely predict the AGM response.

In the 4-variable MIMO system comparisons, we reveal that the difference between players is similar to the 6-variable MIMO.

In RQ1 interactive narrative patterns comparison, we identify three pairs of the 6-variable MIMO and two pairs of the 4-variable MIMO as the primary connections, respectively. The predictability of the pairs differs across the various types of interactions. The response outcome of AlphaGo is more easily forecasted than human players in short-term results changes in machine-machine and human-human games. In contrast, the long-term outcome responses of human players are more predictable in human-machine games.

3.4. RQ2 response magnitude comparison - the parallel comparison

The comparison of response magnitude between different responders under a unified scale can reflect the behavioral alignment in interactions.

In Fig. 5, we compare the frequency response magnitudes. We note that for all players, the response of short-term proceeds to long-term proceeds (LTPs-STPs, LTPs-STPs oppo, and LTPs-NPs) spreads a broader scale than that of other pairs. While in pairs of LTPs-STPs and STPs-STPs, AGM shows a higher response in the short-term proceeds responding to pros' long-term and short-term proceeds. In the 4-variable MIMO, there is little difference between the players in Type A and Type C games. While in Type B games, AGM exhibits a higher response to pros' signals of long-term and net proceeds change (NPs-NPs and LTPs-NPs) than pros do.

Then, we conduct the paired test on the boundaries of a 95% confidence interval of the frequency response, as illustrated in Fig. 6. This statistical handling is an alternative to ensure the "average" interrogator judgment. We focus on the distinct steady difference in comparison.

In the 6-variable MIMO (panel a in Fig. 6), there is little difference in Type C games for both players, which indicates human players have a close response range. Regarding Type B games, the upper boundary for STPs-STPs and LTPs-STPs is positive, indicating that AGM tends to over-react to the signals of pros' short-term and long-term proceeds compared to pros in the same situation. For STPs oppo-STPs oppo and LTPs-STPs oppo, the upper boundary difference is negative for the games between AGM and the pros. It implies that the AGM does not respond as much as the pros in impacting the opponent's STPs corresponding to the change of the opponent's short-term and long-term proceeds on AGM. For STPs-LTPs and STPs oppo-LTPs, the response difference of the upper boundary is around zero but significant in the paired test for the games between AGM and the pros. It also implies that AGM has a significantly different response in upper boundary from pros in the long-term proceeds' response corresponding to the opponent's short-term proceeds. A similar pattern is in Type A games.

In the 4-variable MIMO paired T-test (panel b in Fig. 6), there is some local difference for LTPs-NPs and NPs-NPs in Type C games. On the other hand, the response difference is not as evident as in the 6-variable MIMO in Type A games, but only some local difference in LTPs-LTPs and NPs-NPs. Moreover, in Type B games, AGM seems to have a higher and upper response limit than pros in NPs-NPs and LTPs-NPs. There is also some local difference in the lower boundary in pairs LTPs-LTPs and NPs-LTPs, although this difference approaches zero.

The distinct steady difference in Fig. 6 might indicate the deliberate response difference, but it requires some examination of the mental drive of behavior patterns in future studies. A related surging topic is studying the prospection and metacognition of human behaviors [42]. Meanwhile, the difference in the 6-variable MIMO and the 4-variable MIMO comparison advises that the behavior patterns emerged when the decision signals decomposed more specifically to the impact on different agents. In other words, there is a mental accounting difference between AlphaGo and pros behaviors.

To summarize, we find a different pattern between AlphaGo and human players in RQ2 about the response magnitude comparison. By com-

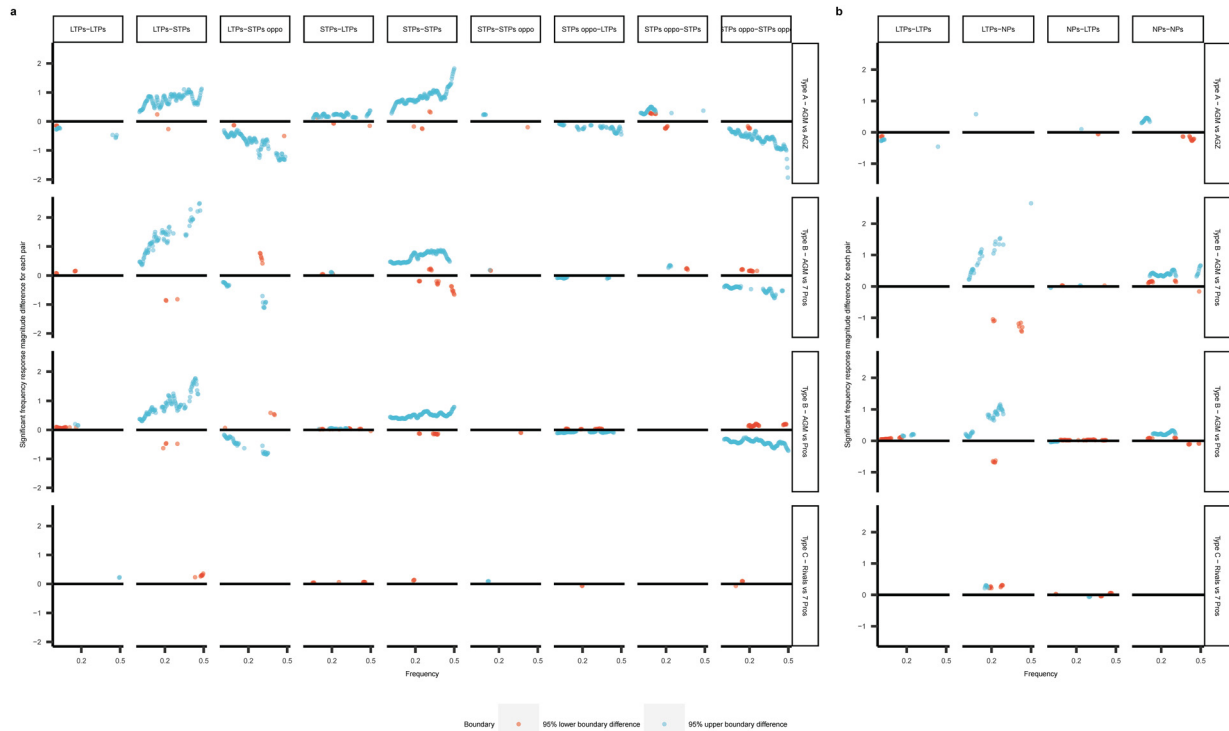


Fig. 6. Comparison of significant paired T-test on frequency response 95% boundaries for 6-variable MIMO (left) and 4-variable MIMO (right). The tags on the top are the stimulus-response pairs. The response difference calculation for different types of games is as below. For Type A. games, it is the average AGM response minus AGZ response for each pair. For Type B. games, it is the average AGM response minus the pros' response for each pair. For Type C. games, it is the average of each pair that rivals' response minus the pros' response. The significance level of the paired T-test: 5%.

paring to human professional players, AlphaGo seems to overreact to its short-term result change but underreact to the opponent's short-term result change.

3.5. RQ3 interaction relationship

We consider that the gain and loss status in interactions reflects the relationship between different players, i.e., mutual benefit or conflict of interest. Hence, we take the interaction relationship to reflect the social dilemma situation in sequential decision interactions.

Since we consider the intertemporal decision as a motion process with gain and loss representing the speed of payoff change, the phase on gain and loss sequence of players in the same game can reflect the dynamic relationship in interactions [23]. Therefore, we interpret the change sequences' co-movement and counter-movement as non-conflict and conflict status, respectively. Likewise, leading or lag status can also be distinguished from the phase. Nevertheless, we have restructured each game into two sub-systems where players act as responders, respectively, which might impact the leading and lag analysis. Therefore, we only focus on conflict and non-conflict relationships in this section.

In Fig. 7, the pair of LTPs-LTPs for the 6-variable MIMO gradually directs to a conflict status (around -180 or 180°) at the low-frequency level. It is easy to understand since the sum of the winning rates for both players is always 100%. There is a competition for the final result in the game. The pairs of STPs oppo-STPs and STPs-STPs oppo both lead to the identical direction (around 0°) at the low-frequency level. It is related to the nature of the game of Go. At the beginning of the game, only a few stones are on the board. One player's decision (STPs) may barely influence the opponent (STPs-oppo). As the game continues, the space on the board shrinks. The player's impact on the opponent gradually increases when the frequency moves to a low level. Therefore, near the end of the game, the moving directions of STPs and STPs-oppo fully agree. In the 4-variable MIMO, LTPs-LTPs and NPs-NPs gradually

change to conflict status as the frequency moves to the low level. The reason is similar to the above in the 6-variable MIMO.

To summarize, we find an identical direction of interactive relationships for all types of games in RQ3 about the social dilemma analysis. The relationship moves from a neutral pattern in the short term to a conflict or identical connection in the long term.

From the studying on RQ1 and RQ3, we note that the three primary pairs identified in the vertical comparison for the 6-variable MIMO and two primary pairs for the 4-variable MIMO have significant statistical directions in the long term, i.e., LTPs-LTPs of the 6-variable MIMO, and LTPs-LTPs, NPs-NPs of the 4-variable MIMO move to the conflict direction in the long term. In contrast, STPs-STPs oppo, STPs oppo-STPs of the 6-variable MIMO move in the identical direction. At the same time, the response magnitude difference is not necessarily significant for all the pairs.

4. Discussion

4.1. Extensions of the empirical results

Several points need to be addressed for the empirical results of AlphaGo, which might help in understanding human decisions in future studies. First, for individual players' narrative features, the advantage of AlphaGo in the long-term outcome change perception and response might relate to the machine's strong capacity in calculation and memory. Second, for the interactive narrative patterns, contrary to intuitions, the response outcome of AlphaGo to the opponent's behaviors can be more easily predicted than humans in short-term outcomes when considering machine-machine games and human-human games. It seems human tends to make more ineffective decisions. Besides the capacity difference, the psychology of humans has an impact on human decisions. For example, pros might pursue making the situation more complex and seek the chance when the opponent makes mistakes. In contrast, the

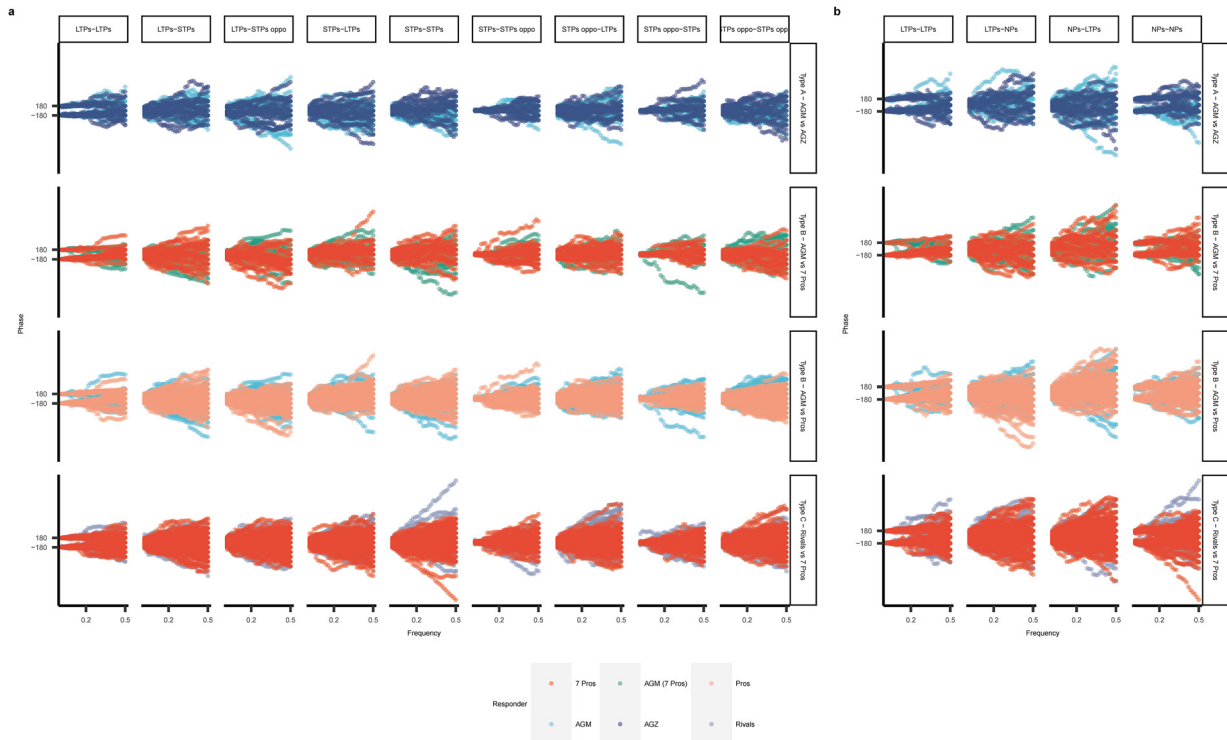


Fig. 7. Phase comparison in different games for the 6-variable MIMO (left) and the 4-variable MIMO (right). For easy comparison, we omit the confidence interval of phase in the charts. The tags on the top are the stimulus-response pairs.

outcome responses to long-term results changes are consistently more predictable for humans than for AlphaGo in human-machine games.

Considering the above individual and interactive empirical narrative patterns, humans might be less sensitive to changes in long-term results than AlphaGo. Instead, humans might concentrate on short-term results and changes in strategic planning.

Then, according to the response magnitude comparison, the overreaction of AlphaGo in its payoff change (STPs) to other signals and underreaction in the opponent payoff change (STPs-oppo) to other signals might imply a deviation in human's response in interactions from optimized patterns (Consider the game results between AlphaGo and human players, AlphaGo's response might be thought of a better pattern). Moreover, the deviation of humans might be rooted in the attention or risk perception difference from the objective patterns of AI.

Finally, the interactive relationship reflects some evolutionary trends of social dilemmas in the game. Social dilemma analysis is not a straightforward or common aspect that traditional HMI research focuses on. Since statistical learning is critical in knowledge acquisition for AI and humans, behavior analysis cannot ignore the significant patterns of interactive relationships. The conflict relationship patterns might help scientists acquire in-depth knowledge about the driven social dilemma and further promote the research related to AI ethics management. The identical relationship patterns suggest some latent mechanisms impacting agent cooperation. From the economic point of view, in a competitive context, behaviors resulting in mutual benefits seem only to exist in the short term (at a high-frequency level). The relationship change might relate to the resource status and the agent's perception of the opponent's strategy. The interactive relationship looks like an even distribution without trends in a non-competitive context.

4.2. Theoretical implication and contribution of the framework

The framework suggests comparing agents' behavior from a novel angle on several dimensions of mind perceptions [43] to judge the difference between humans and machines in HMI. To bridge the deci-

sions and responses in HMI coherently, we suggest MIMO techniques for easy implementation in understanding machine behaviors. The narrative feature can be better observed from a frequency-domain view than the canonical time-domain angle. The contributions of the framework are in four aspects. Firstly, due to the complexity of human decisions, models with pre-defined variables might hide the critical concerns of humans in decision-making. With data-driven approaches, the major structures in HMI will naturally emerge. Therefore, the framework and methodologies suit HMI's complex and less transparent problems. Moreover, the decision patterns of AIs might help scientists notice the blind area for even human experts [44,45]. Secondly, sequential decision data embodies the mental activities of human beings, e.g., perception and memory. Furthermore, the "cognition" of intelligent machines can be understood as analogical. Therefore, the framework based on decision data might interpret the motivations of machine behaviors quickly and clearly. Thirdly, the adaption of AI in HMI is a concern in human-machine cooperation and AI ethics [46]. The empirical comparison explicates the difference between machines and humans in response magnitudes. Moreover, the social dilemma measure might help recognize unethical behaviors. Last, our study extends the canonical decision theory, which traditionally focuses on individual decision patterns, to an interactive context involving multiple factors. In particular, the significant statistical patterns in stimulus-response pairs and interactive relationships imply some critical factors impacting decision-making in the interactive context. The comparison between different types of interactions might shed light on the concept and quantitative measure of the boundary of rationality. Furthermore, the extensions to decision theory in an interactive context help AI technique improvement and novel direction development in HMI analysis.

4.3. Future direction on theoretical analysis and practice

The proposed framework might suggest a possible path to understanding human emotions. Scientists have implemented various exami-

nations in studying sentiment, for instance, facial expression [1,2], physiological signals [5], or text mining [47]. The mechanism of emotion for robots with inner activities is still under discussion [22]. The prospect theory [20] suggests that human emotional changes relate to short-term outcomes of gain and loss. Scientists need empirical exploration to capture the connection between emotion and intertemporal gain and loss. HMI studies with decision proxies might provide a more direct and essential explanation of the emotion mechanism from a psychological perspective.

5. Conclusion

Motivated by the discussion about response-dependence theory in the imitation game, we propose a framework to analyze and compare agents' behavior patterns in HMI with cross-disciplinary knowledge and techniques. From a decision perspective, we put forward the implication of change in agents' interaction and incorporate stimulus-response comparison in behavior pattern analysis. We raise three research questions and apply corresponding multivariate techniques to compare agents' behaviors from static and dynamic perspectives.

Then we conduct an empirical analysis on a real HMI example of the game of Go. Regarding RQ1 about the narrative natures of intertemporal decisions, we note significant differences in the individual cyclical patterns between AI and human professional players. We suggest that AI is more sensitive to long-term results change than humans. Moreover, the interactive narrative pattern proves similar results. For dynamic analysis, vertical comparison brings the primary stimulus-response connectivity to emerge. Regarding RQ2 about the response magnitudes, we reveal AlphaGo overreacts in its short-term payoff change but underreacts to the opponent's short-term payoff change when comparing its performance to professional players. Parallel comparison can illustrate the difference in response magnitude. Regarding RQ3 about the direction of interactive relationships, we determine the obvious statistical patterns of interactive relationships in social dilemma analysis, where the relationship between players gradually moves to a conflict or identical connection for specific connections for all types of games.

It is worth mentioning that comparing the same type of data from several different competent machines is better to ensure robustness. Meanwhile, the experiment on HMI needs to bear a well-defined stimulus-response pair between agents.

Declaration of competing interest

The authors declare that they have no conflicts of interest in this work.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.fmre.2023.12.021](https://doi.org/10.1016/j.fmre.2023.12.021).

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Author profile

Ping Zhao is pursuing a Ph.D. in management science and engineering with the Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China. Her research interests include behavior analysis, complex systems, and economics.

Yunjie Wei received her PhD degree in management science and engineering at Academy of Mathematics and Systems Science, Chinese Academy of Sciences, China in 2017 and also received a PhD degree in management science at City University of Hong Kong in 2018. She is currently an associate professor at Academy of Mathematics and Systems Science, Chinese Academy of Sciences, China. Her research interests include economic modeling, analysis and forecasting. She has published over 20 papers in journals including *IEEE Transactions on Systems, Man and Cybernetics: Systems, Applied Energy*, and *Tourism Management*.

Shouyang Wang received his PhD degree in operations research from the Institute of Systems Science, Chinese Academy of Sciences, China, in 1986. He is currently a Bairen Distinguished Professor of Management Science at the Academy of Mathematics and Systems Science, Chinese Academy of Sciences. He has received any research-related awards and honors. He has published 35 monographs and published more than 330 papers in international academic journals. He is/was a co-editor of 16 journals and a guest editor of special issues/volumes of more than 15 journals. His research interests include decision analysis, risk management, economic analysis and forecasting.